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Angular 20

What are the challenges in modern web development?

1. Unified UX (user experience) :-

- Application must have same experience across all devices.
- Mobile users must get access to everything.

2. Fluid UX :-

- User must stay on one page and get access to everything on to the page.
- New details are added to page without reloading the page.

3. Loosely Coupled And Extensible:-

- Developer can add new features into application without re-installing the complete application.
- New features should not lead to catastrophic failures. (A catastrophic failure is a **sudden and total failure from which recovery is impossible**. Catastrophic failures often lead to cascading systems failure.)

4. Simplified Deployment :-

- Without restarting the device new application should register - Install and open.

What is problem with JavaScript and JQuery

- Lot of Dom manipulations
- Lot of coding - Heavy application
- Hard to test - Hard to extend
- Not support OOP features.

What is Angular?

- Angular is open source, cross platform framework maintained by google and a large community of developers.
- Angular is an developers platform.
- Developers platform provides complete end to end solution for developer.

i.e. from build, debugging, testing to deploying.

- Platform for building SPA and PWA. [Progressive Web Application] (an app that's built using web platform technologies, but that provides a user experience like that of a platform-specific app)

Drawbacks of angular js :

- Google started Angular.js – 2010
- JavaScript based
- Not completely OOP
- Extensibility issues
- Legacy library
- Slow in rendering
- Heavy
- Lot of gaps
- Depend on lot of 3rd party

Features of Angular :-

- Modular Library
- Fast [10x faster than angular.js]
- Platform for developer
- Reduces the integration of 3rd party
- End to End solution for developer [Building to Deploying]
- It uses frameworks like MVC, MVVM [Handling Data Bindings]

Features of Angular :-

1. Developer platform

2. Modular Library

- Application specific framework.
- Faster
- Light weight
- Suitable for low bandwidth devices (limited amount of data can be transmitted over an Internet connection in a given period. It results in sluggish Internet speeds, making performing tasks such as streaming videos or downloading

large files difficult)

3. Differential Loading :-

- Reduces the compatibility issues
- Loads legacy library for legacy browsers and loads modern library for modern browsers

4. Asynchronous :-

- Implicitly uses Ajax (Asynchronous JavaScript and xml) for various interactions
- Improves the performance of application.

Road Map of Angular

Project structure : About files or folder structure

Components : Angular is component based framework. It is class based component. Web page is designed as components. Each partition of the webpage is called Component.

Data binding - To show html data in web page , linking typescript file with html file is called data binding

Component communication : communication between components. to send data between components

Directives : using directives can write code in html page like using if, switch case, loops, etc

Routing : means moving from one component to another component

Forms :

Template-Driven Approach

Reactive Approach

Pipes : It is also called filters. It takes input in one format and gives output in another format. Let's say, date is in one format. to show date in another format then pipes is used. predefined pipes are existing.

Services : It is used to send same data to multiple components. It contains the code used to access data from database.

Http : using this http module you can send request to database or api

Setup Environment for Angular :-

- Download and Install Node.js for Package Manager : NPM

Nodejs is used to convert type script to javascript.

<https://nodejs.org> -> click on - download Node js (LTS – long term support)

In downloads folder -> dbl click on - node-v20.20.0-x64 -> next -> next -> finish.

To get node js version:

```
C:\> node -v
```

v20.20.0

2. Download and Install TypeScript

```
C:\> npm install -g typescript
```

Note:

To install typescript specific version.

```
C:\> npm install -g typescript@5.9.0
```

To see the Version:

```
C:\> tsc -v
```

Version 5.9.0

- To uninstall typescript

```
C:\> npm uninstall -g typescript
```

3. Download and Install Visual Studio Code Editor

4. Download and Install Angular CLI

[CLI is a command line tool used for creating and managing angular applications]

```
C:\> npm install @angular/cli -g
```

```
C:\> ng --version
```

ng means angular command.

Update from older version to latest :-

1. un-Install the existing version

```
C:\> npm uninstall -g @angular/cli
```

2. Clear Cache

```
C:\> npm cache verify
```

3. Install Latest version

```
C:\>npm install -g @angular/cli
```

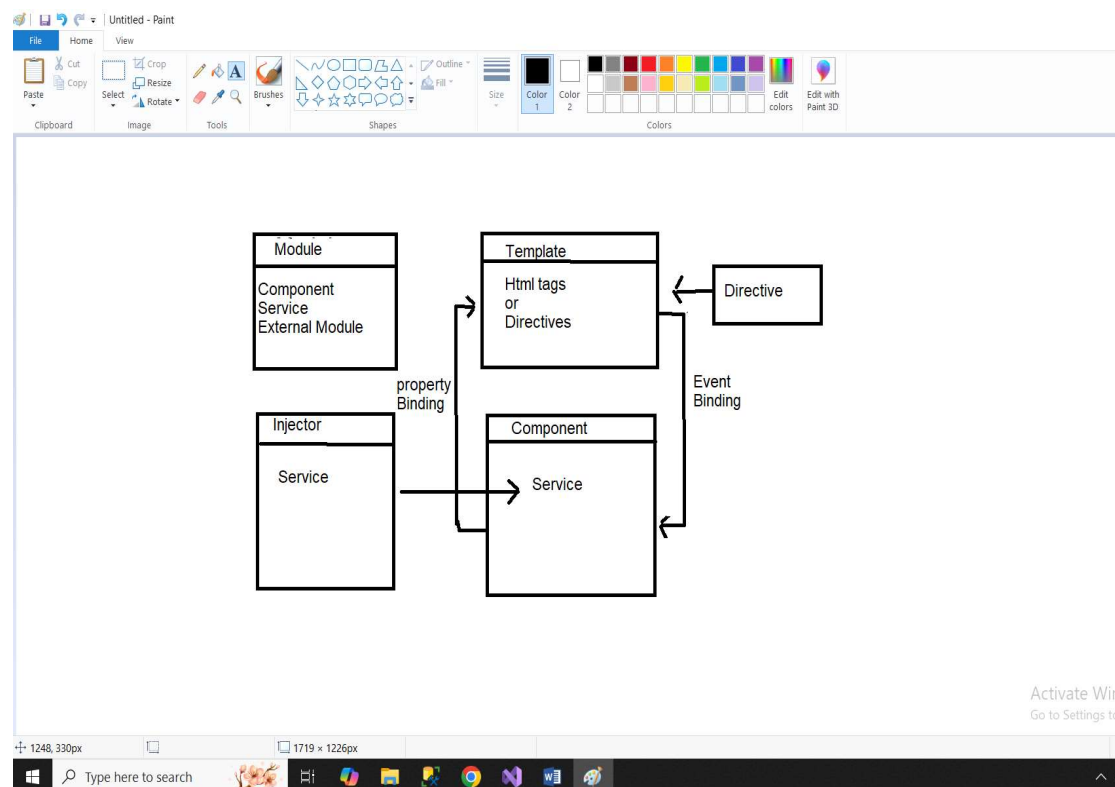
-g (globally) angular works from any location

Note :- For any specific angular version

```
D:\>npm install -g @angular/cli@20
```

angular.io - version numbers

Angular Architecture



Angular Application Flow :-

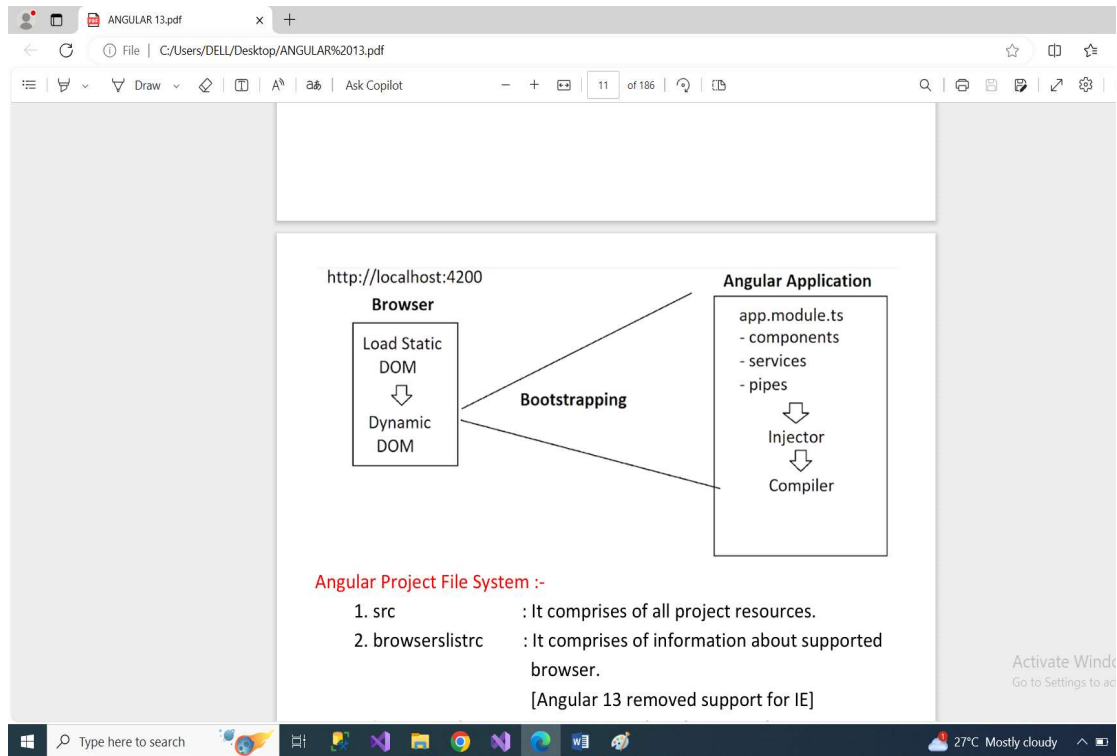
1. You compile angular application
2. "app.module.ts" identifies all the library required for angular application and injects into memory.
3. Babel uses AOT mechanism and converts that static DOM into dynamic DOM.

The Angular ahead-of-time (AOT) compiler converts your Angular HTML(Angular forms) and TypeScript code into efficient JavaScript code during the build phase before the browser downloads and runs that code. Compiling your application during the build process provides a faster rendering in the browser

Babel is a popular tool that is mainly used to convert ECMAScript 2015+ code

(European Computer Manufacturers Association) into a backward-compatible version of JavaScript in current and older browsers or environments. It can also convert TypeScript code into JavaScript.

4. The process of converting static DOM into dynamic DOM is known as "bootstrapping".



Folder structure of Angular Application

node_modules

src

app

assets

favicon.ico

index.html

main.ts

styles.css

.editorconfig

.gitignore

angular.json

package-lock.json

package.json

tsconfig.app.json

tsconfig.json

tsconfig.spec.json

- `node_modules` : It contains of all library files installed through NPM.
- `src` : It contains of all project resources.
 - `app` : It contains of application resources like modules, components, services, pipes etc
 - `assets` : It contains of static resources like images, - text documents, pdf, word, audio, video etc.
 - `favicon.ico` : It is the favourite icon for website.
 - `Index.html` : It is the startup page
 - `main.ts` : Entry point, from here your application starts compiling, processing.
 - `styles.css` : It contains of global styles. Styles used for all components.
- `.editorconfig` : It contains of rules for editor used by developers.
- `.gitignore` : It contains of configuration required for publishing on GIT.
- `angular.json` : It configures the initial environment for Projects created in angular.
- `package-lock.json` : It contains of meta data about every dependency installed for project.
- `package.json` : It contains of meta data.
- `tsconfig.app.json` : It defines the rules for typescript.
- `tsconfig.json` : It configures the rules for typescript. It also defines the environment for typescript in angular application.
- `tsconfig.spec.json` : 'spec' refers to testing file.